

## SAFETY DATA SHEET

Creation Date 03-Jan-2005

Revision Date 24-Dec-2021

Revision Number 5

### 1. Identification

**Product Name** Diisobutylaluminium hydride, 1.1M solution in cyclohexane

**Cat No. :** AC201030000; AC201031000; AC201038000

**Synonyms** DIBAL-H, 1.1M solution in cyclohexane

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                        |              |
|------------------------------------------------------------------------|--------------|
| Flammable liquids                                                      | Category 2   |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1   |
| Skin Corrosion/Irritation                                              | Category 1 A |
| Serious Eye Damage/Eye Irritation                                      | Category 1   |
| Specific target organ toxicity (single exposure)                       | Category 3   |
| Target Organs - Central nervous system (CNS), Respiratory system.      |              |
| Aspiration Toxicity                                                    | Category 1   |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor

In contact with water releases flammable gases which may ignite spontaneously  
May be fatal if swallowed and enters airways  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May cause drowsiness or dizziness



### **Precautionary Statements**

#### **Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep away from any possible contact with water, because of violent reaction and possible flash fire  
Handle under inert gas. Protect from moisture  
Keep cool

#### **Response**

Immediately call a POISON CENTER or doctor/physician

#### **Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

#### **Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse  
Brush off loose particles from skin. Immerse in cool water/wrap with wet bandages

#### **Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

#### **Ingestion**

Do NOT induce vomiting  
Rinse mouth

#### **Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

#### **Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in a dry place. Store in a closed container

#### **Disposal**

Dispose of contents/container to an approved waste disposal plant

#### **Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects  
Reacts violently with water

## **3. Composition/Information on Ingredients**

| Component                  | CAS No    | Weight % |
|----------------------------|-----------|----------|
| Cyclohexane                | 110-82-7  | 80       |
| Diisobutylaluminum hydride | 1191-15-7 | 20       |

#### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                             |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                             |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                                                                                       |
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration).                                                           |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                         |

#### 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | DO NOT USE WATER                                                                                                    |
| <b>Flash Point</b>                      | -18 °C / -0.4 °F                                                                                                    |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Burning produces obnoxious and toxic fumes. Isobutane.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

Health  
3

Flammability  
3

Instability  
2

Physical hazards  
W

## 6. Accidental release measures

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods for Containment and Clean Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

### Handling

Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Storage.

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep from any possible contact with water. Corrosives area. Flammables area. Keep under nitrogen. Keep away from water or moist air. Incompatible Materials. Acids. Water. Strong oxidizing agents. Alcohols. oxygen.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                   | ACGIH TLV    | OSHA PEL                                                                                                       | NIOSH IDLH                                                    | Mexico OEL (TWA) |
|-----------------------------|--------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------|
| Cyclohexane                 | TWA: 100 ppm | (Vacated) TWA: 300 ppm<br>(Vacated) TWA: 1050 mg/m <sup>3</sup><br>TWA: 300 ppm<br>TWA: 1050 mg/m <sup>3</sup> | IDLH: 1300 ppm<br>TWA: 300 ppm<br>TWA: 1050 mg/m <sup>3</sup> | TWA: 100 ppm     |
| Diisobutylaluminium hydride |              | (Vacated) TWA: 2 mg/m <sup>3</sup>                                                                             | TWA: 2 mg/m <sup>3</sup>                                      |                  |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                                   |
|-----------------------------------------------|-----------------------------------|
| <b>Physical State</b>                         | Liquid                            |
| <b>Appearance</b>                             | Colorless                         |
| <b>Odor</b>                                   | sweet                             |
| <b>Odor Threshold</b>                         | No information available          |
| <b>pH</b>                                     | Not applicable                    |
| <b>Melting Point/Range</b>                    | -70 °C / -94 °F                   |
| <b>Boiling Point/Range</b>                    | No information available          |
| <b>Flash Point</b>                            | -18 °C / -0.4 °F                  |
| <b>Evaporation Rate</b>                       | No information available          |
| <b>Flammability (solid,gas)</b>               | Not applicable                    |
| <b>Flammability or explosive limits</b>       |                                   |
| Upper                                         | No data available                 |
| Lower                                         | No data available                 |
| <b>Vapor Pressure</b>                         | No information available          |
| <b>Vapor Density</b>                          | No information available          |
| <b>Specific Gravity</b>                       | 0.780                             |
| <b>Solubility</b>                             | Reacts violently with water       |
| <b>Partition coefficient; n-octanol/water</b> | No data available                 |
| <b>Autoignition Temperature</b>               | No information available          |
| <b>Decomposition Temperature</b>              | No information available          |
| <b>Viscosity</b>                              | No information available          |
| <b>Molecular Formula</b>                      | C <sub>8</sub> H <sub>19</sub> Al |
| <b>Molecular Weight</b>                       | 142.22                            |

## 10. Stability and reactivity

|                                         |                                                                                                                                                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                                                                                                                  |
| <b>Stability</b>                        | Reacts violently with water. Moisture sensitive. Air sensitive.                                                                                                                                      |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. Exposure to air. Do not allow evaporation to dryness. Incompatible products. Exposure to moist air or water. Exposure to moisture. |
| <b>Incompatible Materials</b>           | Acids, Water, Strong oxidizing agents, Alcohols, oxygen                                                                                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Burning produces obnoxious and toxic fumes, Isobutane                                                                                       |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                                                                                                            |
| <b>Hazardous Reactions</b>              | Reacts violently with water.                                                                                                                                                                         |

## 11. Toxicological information

## Acute Toxicity

### Product Information

#### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

#### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

#### Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

### Component Information

| Component   | LD50 Oral            | LD50 Dermal             | LC50 Inhalation                            |
|-------------|----------------------|-------------------------|--------------------------------------------|
| Cyclohexane | > 5000 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | LC50 > 32880 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic** No information available

### Products

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                  | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------|-----------|------------|------------|------------|------------|------------|
| Cyclohexane                | 110-82-7  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Diisobutylaluminum hydride | 1191-15-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS) Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** Category 1

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component   | Freshwater Algae   | Freshwater Fish                                                                                                                                             | Microtox                                        | Water Flea          |
|-------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------|
| Cyclohexane | EC50 >500 mg/L/72h | LC50: 48.87 - 68.76 mg/L,<br>96h static (Poecilia reticulata)<br>LC50: 24.99 - 44.69 mg/L,<br>96h static (Lepomis macrochirus)<br>LC50: 23.03 - 42.07 mg/L, | EC50 = 85.5 mg/L 5 min<br>EC50 = 93 mg/L 10 min | EC50 = 0.9 mg/l/48h |

|  |  |                                                                                                    |  |  |
|--|--|----------------------------------------------------------------------------------------------------|--|--|
|  |  | 96h static (Pimephales promelas)<br>LC50: 3.96 - 5.18 mg/L, 96h flow-through (Pimephales promelas) |  |  |
|--|--|----------------------------------------------------------------------------------------------------|--|--|

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** .

| Component   | log Pow |
|-------------|---------|
| Cyclohexane | 3.44    |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component              | RCRA - U Series Wastes | RCRA - P Series Wastes |
|------------------------|------------------------|------------------------|
| Cyclohexane - 110-82-7 | U056                   | -                      |

### 14. Transport information

#### DOT

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Technical Name (DIISOBUTYLALUMINUM HYDRIDE, CYCLOHEXANE)  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3  
 Packing Group I

#### TDG

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3, 8  
 Packing Group I

#### IATA

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3  
 Packing Group I

#### IMDG/IMO

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3  
 Packing Group I

### 15. Regulatory information

#### United States of America Inventory

| Component                  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Cyclohexane                | 110-82-7  | X    | ACTIVE                                        | -                           |
| Diisobutylaluminum hydride | 1191-15-7 | X    | ACTIVE                                        | -                           |

**Diisobutylaluminium hydride, 1.1M solution in cyclohexane**

Revision Date 24-Dec-2021

**Legend:****TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                  | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Cyclohexane                | 110-82-7  | X   | -    | 203-806-2 | X     | X    | X    | X    | X     | KE-18562 |
| Diisobutylaluminum hydride | 1191-15-7 | -   | X    | 214-729-9 | X     | X    | X    | X    | X     | KE-10903 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

| Component   | CAS No   | Weight % | SARA 313 - Threshold Values % |
|-------------|----------|----------|-------------------------------|
| Cyclohexane | 110-82-7 | 80       | 1.0                           |

**SARA 311/312 Hazard Categories** See section 2 for more information**CWA (Clean Water Act)**

| Component   | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Cyclohexane | X                          | 1000 lb                     | -                      | -                         |

**Clean Air Act** Not applicable**OSHA** - Occupational Safety and Health Administration Not applicable

| Component                  | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|----------------------------|----------------------------------|----------------------------|
| Diisobutylaluminum hydride | -                                | TQ: 5000 lb                |

**CERCLA** This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component   | Hazardous Substances RQs | CERCLA EHS RQs |
|-------------|--------------------------|----------------|
| Cyclohexane | 1000 lb                  | -              |

**California Proposition 65** This product does not contain any Proposition 65 chemicals.**U.S. State Right-to-Know Regulations**

| Component                  | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------|---------------|------------|--------------|----------|--------------|
| Cyclohexane                | X             | X          | X            | -        | X            |
| Diisobutylaluminum hydride | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component                  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cyclohexane                | -                                                                   | Use restricted. See item 57.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |
| Diisobutylaluminum hydride | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)                                                                       | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                  | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Cyclohexane                | 110-82-7  | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Diisobutylaluminum hydride | 1191-15-7 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cyclohexane                | 110-82-7  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Diisobutylaluminum hydride | 1191-15-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By**

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**Creation Date**

03-Jan-2005

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**Print Date**

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**Revision Summary**

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**